Optimizing Edge-AI for 6G via Split and Federated Learning Paradigms

Master program(s): M(I)EET, M(I)ECT, MEI, MEA
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Area: Telecommunications
Keywords: 6G, Machine Learning, Edge Computing, Digital Signal Processing, FPGA, SoC

Scope and Motivation

The evolution toward sixth-generation (6G) networks demands highly efficient, privacy-preserving, artificial intelligence (AI)-based processing directly at the network edge. Traditional centralized machine learning (ML) approaches struggle to meet these requirements due to bandwidth limitations, privacy concerns, and the massive scale of connected devices. Distributed learning (DL) is a promising paradigm that enables collaborative model training across edge devices without sharing raw data, thereby enhancing privacy and reducing communication overhead. Field-programmable gate arrays (FPGAs) support this by offering reconfigurable, heterogeneous, and custom design solutions well suited to edge requirements. This dissertation will explore novel DL frameworks tailored for 6G edge intelligence, focusing on hardware-aware implementations leveraging FPGAs to meet future wireless networks' stringent latency and energy constraints.

Objectives

- Design split and develop federated learning architectures optimized for 6G edge environments.
- Evaluate the performance of DL models on resource-constrained FPGAs platforms.
- Investigate privacy-preserving techniques and their integration into distributed learning frameworks for secure edge AI.
- Analyze the communication and computation trade-offs between split and federated learning in multi-FPGA setups.
- Propose optimization strategies that enhance scalability and reduce latency in distributed learning systems for next-generation applications.

Workplan

First Semester

- Conduct a review of split and federated learning and their applications in edge 6G computing.
- Identify constraints imposed by 6G environments and FPGA limitations.
- Propose FPGA-based solutions tailored to the identified challenges and requirements.
- Mid-term report and presentation.

Second Semester

- Develop FPGA-based split and federated learning algorithms on representative edge scenarios.
- Evaluate models based on accuracy, communication efficiency, privacy preservation, and latency under various network conditions.
- Deploy an experimental testbed that accounts for varying hardware capabilities to replicate real-world edge application scenarios.
- Final dissertation writing, defense preparation, and oral presentation.